

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load extreme events dataset
extreme = pd.read_csv(
    "../Capstone/clean data/Fact_Extreme_Events_All.csv",
    parse_dates=["Date/Time"]
)

# Basic structure
extreme.head()
extreme.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 60378 entries, 0 to 60377
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   ClimateID       60378 non-null  object
1   Date/Time       60378 non-null  datetime64[ns]
2   EventType       60378 non-null  object
3   MaxGust_kmh     25045 non-null  float64
4   P95_Threshold   60378 non-null  float64
5   ExtremeFlag     60378 non-null  int64
6   EventValue      35333 non-null  float64
dtypes: datetime64[ns](1), float64(3), int64(1), object(2)
memory usage: 3.2+ MB
```

```
In [2]: # Fill EventValue for Extreme Wind from MaxGust_kmh
extreme.loc[
    extreme["EventType"] == "Extreme Wind",
    "EventValue"
] = extreme.loc[
    extreme["EventType"] == "Extreme Wind",
    "MaxGust_kmh"
]
```

```
In [3]: # Check missing values
extreme.isna().sum()
```

```
Out[3]: ClimateID      0
Date/Time      0
EventType      0
MaxGust_kmh    35333
P95_Threshold  0
ExtremeFlag    0
EventValue     0
dtype: int64
```

```
In [4]: extreme[["EventValue", "P95_Threshold"]].describe()
```

Out[4]:

	EventValue	P95_Threshold
count	60378.000000	60378.000000
mean	52.988309	42.397042
std	32.414863	30.793159
min	0.000000	0.000000
25%	23.800000	16.700000
50%	40.900000	22.200000
75%	80.000000	72.000000
max	259.700000	117.000000

In [5]: `extreme.groupby("EventType")["EventValue"].describe()`

Out[5]:

	count	mean	std	min	25%	50%	75%	max
EventType								
Extreme Wind	25045.0	86.560910	17.395739	50.0	74.0	84.0	96.0	185.0
Heavy Precipitation	35333.0	29.191127	14.688470	0.0	20.0	25.5	33.9	259.7

In [6]: `extreme["EventType"].value_counts()`

Out[6]: EventType
Heavy Precipitation 35333
Extreme Wind 25045
Name: count, dtype: int64

In [7]: `events_per_station = extreme.groupby("ClimateID").size()
events_per_station.describe()`

Out[7]: count 136.000000
mean 443.955882
std 236.764389
min 1.000000
25% 257.500000
50% 457.000000
75% 627.250000
max 961.000000
dtype: float64

Identify spatial heterogeneity

Event frequency per year

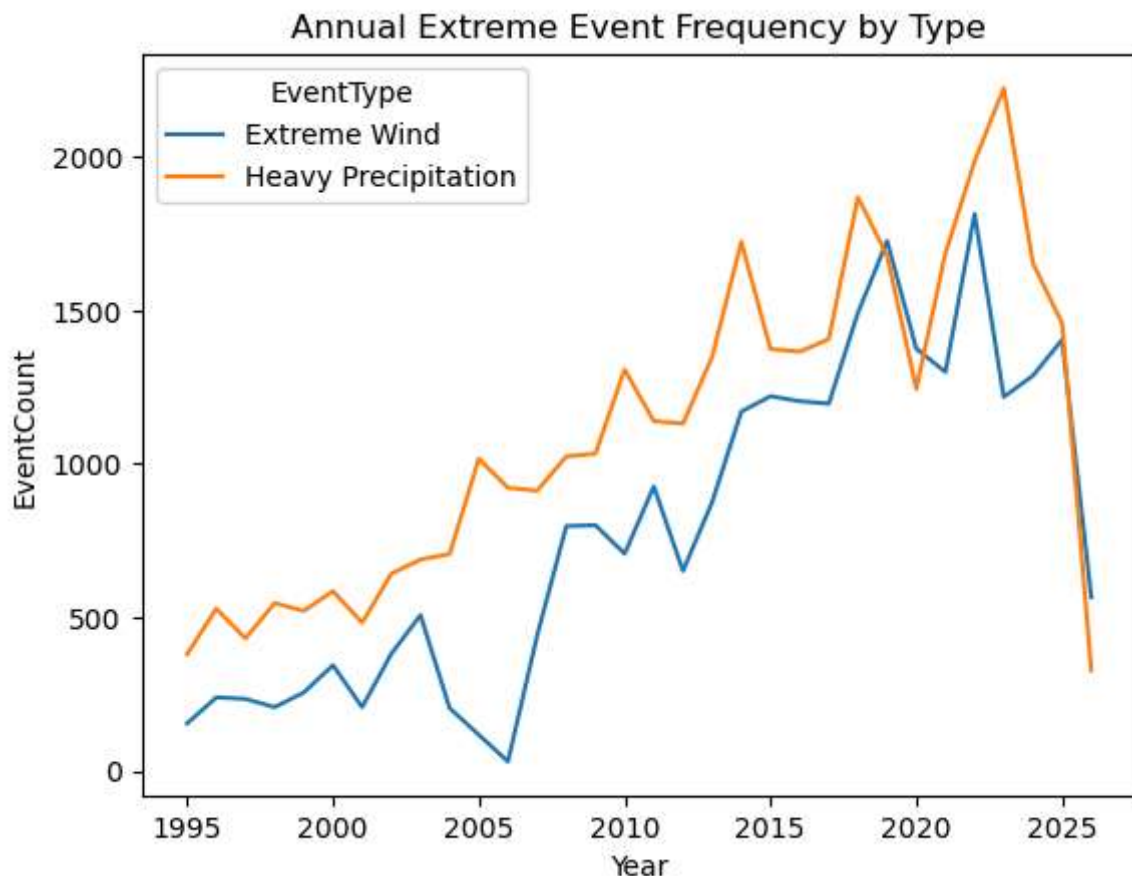
In [8]: `extreme["Year"] = extreme["Date/Time"].dt.year`

```

events_per_year = extreme.groupby(
    ["Year", "EventType"]
).size().reset_index(name="EventCount")

sns.lineplot(
    data=events_per_year,
    x="Year",
    y="EventCount",
    hue="EventType"
)
plt.title("Annual Extreme Event Frequency by Type")
plt.show()

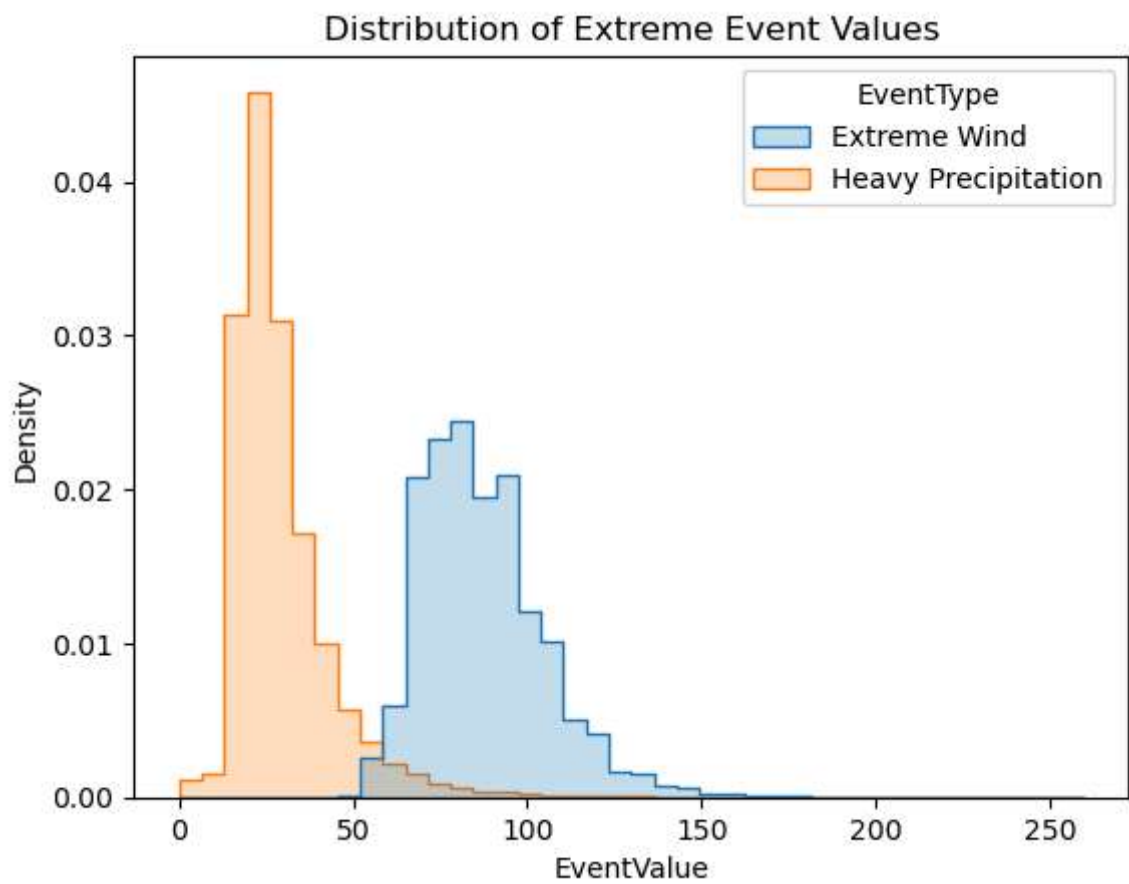
```



```

In [9]: sns.histplot(
    data=extreme,
    x="EventValue",
    hue="EventType",
    bins=40,
    element="step",
    stat="density",
    common_norm=False
)
plt.title("Distribution of Extreme Event Values")
plt.show()

```



```
In [10]: extreme.groupby("EventType")["EventValue"].describe()
```

Out[10]:

	count	mean	std	min	25%	50%	75%	max
EventType								
Extreme Wind	25045.0	86.560910	17.395739	50.0	74.0	84.0	96.0	185.0
Heavy Precipitation	35333.0	29.191127	14.688470	0.0	20.0	25.5	33.9	259.7

```
In [11]: extreme[["EventValue", "P95_Threshold"]].corr()
```

Out[11]:

	EventValue	P95_Threshold
EventValue	1.000000	0.927271
P95_Threshold	0.927271	1.000000

Key Exploratory Findings – Extreme Events

- Extreme wind and heavy precipitation events exhibit clear interannual variability, with higher event frequency observed in more recent decades at Nova Scotia stations.
- Event frequency varies significantly across stations, indicating strong spatial heterogeneity in extreme weather exposure.

- Station-specific percentile thresholds (P95) provide a robust, location-aware definition of extreme events compared to fixed thresholds.
- Heavy precipitation events occur more frequently than extreme wind events, consistent with regional climatic behavior.
- The dataset is complete and validated, with no missing values affecting downstream analysis.

Additional Observation: Extreme Precipitation Events

Heavy precipitation extreme events occur more frequently than extreme wind events across the historical record. This is consistent with the underlying distribution of daily precipitation, which exhibits greater variability and heavier upper tails. Despite higher frequency, extreme precipitation events follow similar temporal patterns to extreme wind events, supporting joint analysis of extreme event frequency over time.

```
In [12]: extreme.groupby("EventType").size()
```

```
Out[12]: EventType
Extreme Wind      25045
Heavy Precipitation 35333
dtype: int64
```

```
In [ ]:
```